FOODRUGS: AN EXPLORATORY MINING WEB APPLICATION FOR FOOD - DRUG INTERACTIONS

P240-T

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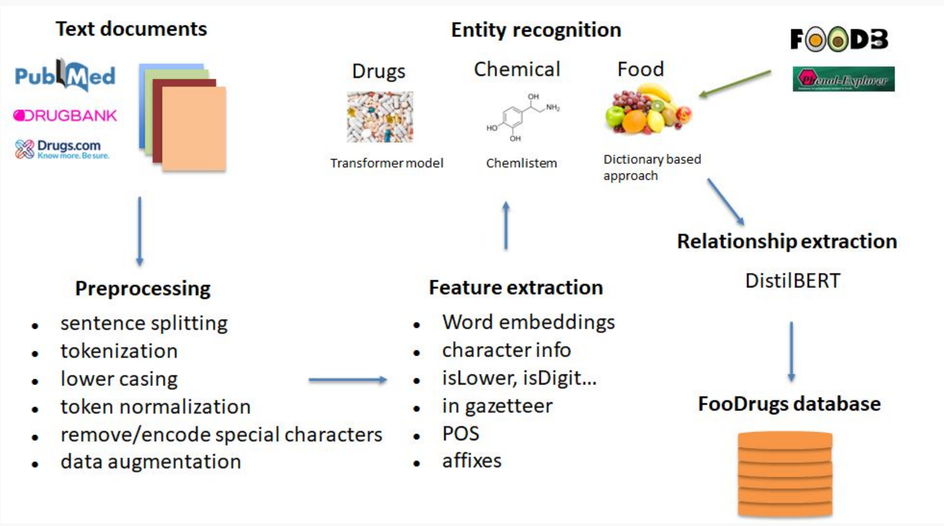
ABSTRACT

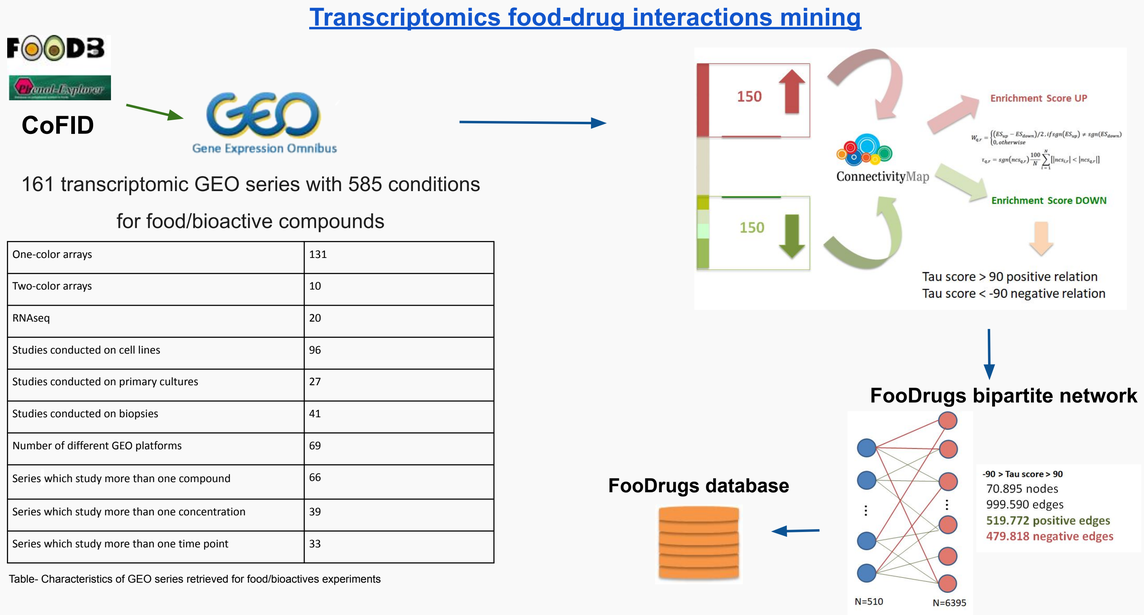
The consumption of supplement foods with an active role to prevent non-communicable diseases have increased during the last years raising concerns over the potential interactions with drugs, particularly in patients undergoing chronic therapy. However, the current knowledge of them in clinical practice has been reported to be unsatisfactory. Therefore, It is required to develop computational tools to exploit the vast information available in dispersed databases about food-drug interactions. We present FOODRUGS, a friendly-user webapp to access and explore putative food-drug interactions. Users can explore transcriptomic relationships between food and drugs through a similarity bipartite network built with 585 food transcriptomic profiles from GEO and the CMAP database for drug profiles. Food and drug nodes are connected by a positive or negative tau score representing the similarity (positive) or dissimilarity (negative) relationship, meaning for a putative similar or opposite mechanism of action. In addition, enrichment analysis to annotate drug compounds for a food node can be performed. Lastly, more than 2.500 scientific documents from Pubmed, drugbank and drugs.com can be consulted after applying NLP techniques to extract relevant food-drug interactions.

FOODRUGS DATABASE

Text documents food-drug interactions mining



Transcriptomics food-drug interactions mining



One-color arrays	131
Two-color arrays	10
RNAseq	20
Studies conducted on cell lines	96
Studies conducted on primary cultures	27
Studies conducted on biopsies	41
Number of different GEO platforms	69
Series which study more than one compound	66
Series which study more than one concentration	39
Series which study more than one time point	33

Table- Characteristics of GEO series retrieved for food/bioactives experiments

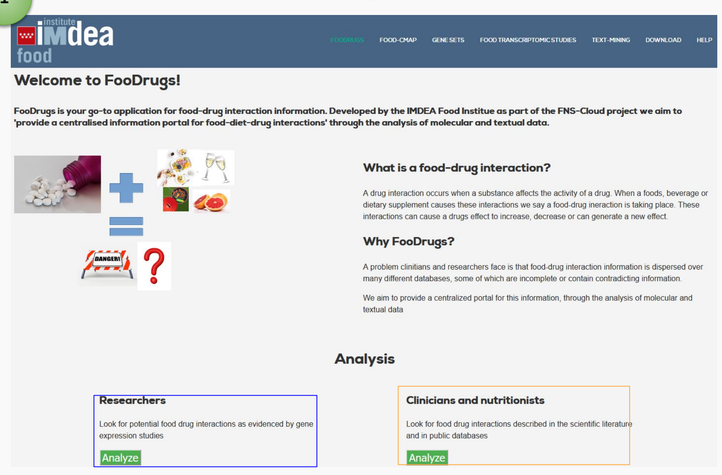
Cite: Carrillo de Santa Pau, Enrique, Garranzo, Marco, & Laguna Lobo, Teresa. (2022). FooDrugs database: A database with molecular and text information about food - drug interactions (2.0.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.6638470

Access FooDrugs database: http://imdeafoodcompubio.com/index.php/download/

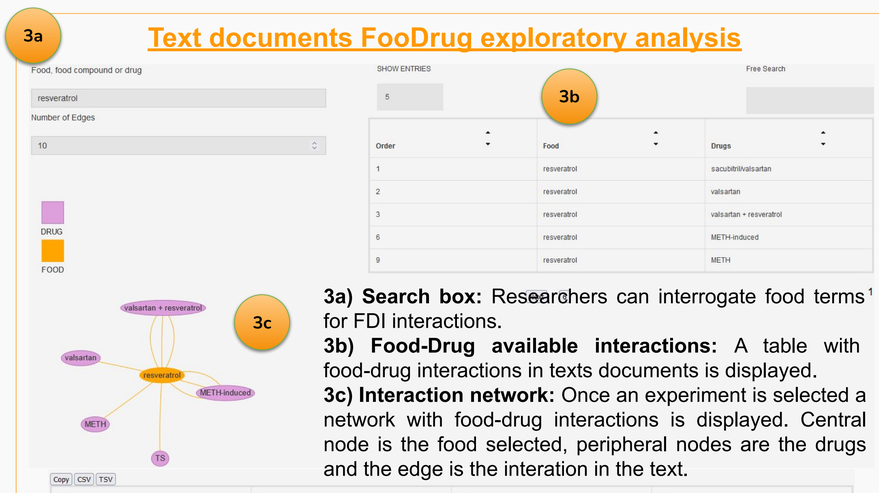
FOODRUGS WEBTOOL

Access: http://imdeafoodcompubio.com/index.php/foodrugs/

1 Welcome website and analysis access



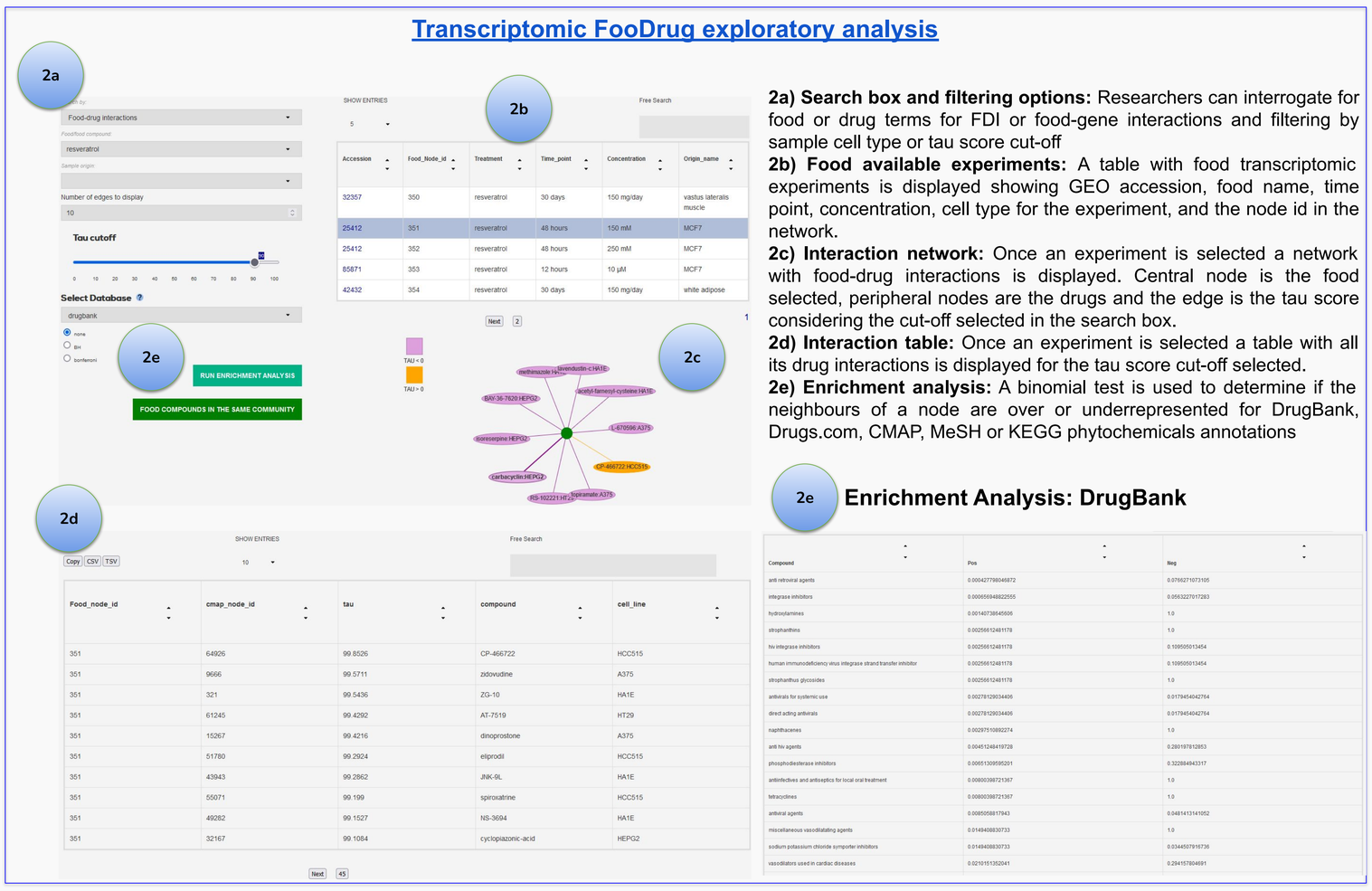
3a Text documents FooDrug exploratory analysis



3a) Search box: Researchers can interrogate food terms for FDI interactions.
3b) Food-Drug available interactions: A table with food-drug interactions in texts documents is displayed.
3c) Interaction network: Once an experiment is selected a network with food-drug interactions is displayed. Central node is the food selected, peripheral nodes are the drugs and the edge is the interation in the text.

3d) Interaction table: Once an experiment is selected a table with the text where the interaction is mention indisplayed. In blue the food and in red the drug,in bold black the mention of the interaction. In addition, we provide the link to the document.

Transcriptomic FooDrug exploratory analysis



2a) Search box and filtering options: Researchers can interrogate for food or drug terms for FDI or food-gene interactions and filtering by sample cell type or tau score cut-off
2b) Food available experiments: A table with food transcriptomic experiments is displayed showing GEO accession, food name, time point, concentration, cell type for the experiment, and the node id in the network.
2c) Interaction network: Once an experiment is selected a network with food-drug interactions is displayed. Central node is the food selected, peripheral nodes are the drugs and the edge is the tau score considering the cut-off selected in the search box.
2d) Interaction table: Once an experiment is selected a table with all its drug interactions is displayed for the tau score cut-off selected.
2e) Enrichment analysis: A binomial test is used to determine if the neighbours of a node are over or underrepresented for DrugBank, Drugs.com, CMAP, MeSH or KEGG phytochemicals annotations

2e Enrichment Analysis: DrugBank

RESOURCES USED

- **DrugBank:** Wishart, David S., et al. 2018. "DrugBank 5.0: A Major Update to the DrugBank Database for 2018." *Nucleic Acids Research* 46 (D1): D1074–82.
- **Phenol-Explorer:** Rothwell, Joseph A., et al. 2013. "Phenol-Explorer 3.0: A Major Update of the Phenol-Explorer Database to Incorporate Data on the Effects of Food Processing on Polyphenol Content." *Database:* 2013 (October): bat070
- **Chemlistem:** Corbett, Peter, and John Boyle. 2018. "Chemlistem: Chemical Named Entity Recognition Using Recurrent Neural Networks." *Journal of Cheminformatics* 10 (1): 59
- **CMAP:** Subramanian, Aravind, Rajiv Narayan, Steven M. Corsello, David D. Peck, Ted E. Natoli, Xiaodong Lu, Joshua Gould, et al. 2017. "A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles." *Cell* 171 (6): 1437–52.e17.
- **PUBMED:** https://pubmed.ncbi.nlm.nih.gov/
- **Drug.com:** https://www.drugs.com/
- **FooDB:** https://foodb.ca/
- **CoFID:** https://fdnc.quadram.ac.uk

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Food Nutrition Security

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